

Agile Experiences in a Software Service Company

Jakub Rudzki
Solita Oy
Satakunnankatu 18 A
33210 Tampere, Finland
Email: jakub.rudzki@solita.fi

Imed Hammouda
Tampere University of Technology
Korkeakoulunkatu 1
33101 Tampere, Finland
Email: imed.hammouda@tut.fi

Tuomas Mikkola
Solita Oy
Satakunnankatu 18 A
33210 Tampere, Finland
Email: tuomas.mikkola@solita.fi

Abstract—Agile practices are used more commonly in commercial projects. The results of such practices are dependent on many internal and external factors in projects and organisations. In this paper we present the key results of a study of 18 commercial projects based on qualitative and quantitative analysis of data collected. As a research methodology, we use the Goal Question Metric approach. Based on the findings, we suggest a set of recommendations that aim at helping projects in similar commercial context to choose best agile practices.

Keywords—agile; scrum; practical experience;

I. INTRODUCTION

Agile practices in commercial projects have sparked much discussion on their applicability and potential benefits. In spite of the large number of reports on practical usage of Agile methods in small [1] and big organisations [2], it seems that there is not a simple recipe how to successfully use Agile in the context of software service companies. Additionally, the term Agile practices is often understood differently by different people [3]. Despite the obscurities, software companies are increasingly taking into use Agile practices in their projects.

An example of such companies is Solita, a software service company which offers different software solutions for their clients. Solita started using Agile practices a few years ago based on reported positive aspects of Agility, which was also in line with the company open culture. Recently, Solita observed a need to verify how the adoption of Agile practices in the company affected its software projects. A study was therefore conducted.

The rest of this paper is structured as follows. In Section II we present existing work and the details of the research methodology used. Our results are presented in Section III. The recommendations formulated based on the results are presented in Section IV. Finally, we conclude in Section V.

II. BACKGROUND AND RESEARCH METHODOLOGY

Agile practices have been studied for some time [4], [5], [6]. The principles of Agility are gathered in the Agile Manifesto [7]. A good overview of different practices classified as Agile was presented by Abrahamsson et al. [8]. From the different Agile methodologies listed we focus only on the Scrum approach [9], [10], [11], as it is the only Agile methodology used in our case company, Solita. There are also experience reports on Agile usage in the

industry. They specifically focus on large organisations [2], [12] or small ones [1] that adopt or consider the adoption of Agile development. In addition to works on the Agile practices, there were studied practices [11], [13], [14], [15] in distributed projects, such projects were also realised at Solita.

The Goal Question Metric (GQM) approach [16], which was the method used in this work, is meant to allow organisations to measure their specific goals through defining questions and relevant metrics for those questions. The GQM approach has been used in the industry for significant period of time. There are experience reports of the method usage [17], [18]. The GQM defines a specific goal, as the case of our study. Furthermore, the data available was considered sufficient for applying GQM.

We have formulated the goal of this research according to the GQM template. Therefore, our research goal is: (*Purpose*) Find out (*Issue*) the impact of (*Object*) Agile practices on software projects (*Viewpoint*) from the software service provider company point of view. This goal was set to verify whether the Agile practices used in projects make them work better. In order to achieve the goal, we asked five questions that take into consideration different aspects of project realisation in a company. The questions were limited to the ones that were the most relevant to the goal set and were possible to answer in the particular context.

Q1: Do the current Agile practices benefit projects?

In this question we wanted to verify whether there are differences between Agile and non-Agile projects. We used three metrics to get the perspectives of different stakeholders: *Average customer satisfaction in Agile and non-Agile projects*, *Average team performance in Agile and non-Agile projects*, and *Average project profitability in Agile and non-Agile projects*. The customer satisfaction should give the viewpoint of external customers. The team performance is the internal point of view seen by the project manager. Finally, the project profitability represents the internal point of view in the whole company.

Q2: Does the customer direct involvement in the project benefit project success? In this question we wanted to find out whether projects in which customer was actively participating performed better than those without customer participation. We measured this based on customer satisfaction, profitability, and team performance with relation to degree of the customer involvement and the methodol-

ogy used.

Q3: Is the communication different in Agile projects? One of reported benefits of Agile practices, particularly Scrum, is the improvement of communication [11]. Therefore we wanted to verify whether there were any differences in communication-related tasks in Agile and non-Agile projects. We used two metrics that aim at answering this question: *Ratio of communication time to all time spent in a project* and *Project manager involvement in Agile and non-Agile projects*. The communication metric should show whether there are any significant differences in the amount of time spent for communication. The additional metric measuring the time spent for project management tasks aims at showing whether the project manager effort in Agile and non-Agile projects differs.

Q4: How to adapt Agile practices to suite commercial needs? The Agile practices include many different elements that in ideal cases should be used as a whole. However, in commercial projects often there are limits to what can be used. Therefore, we wanted to find out what Agile elements were commonly used in commercial projects and how the projects coped with possible limitations imposed by the external constraints. In order to answer this question we used qualitative data collected from interviews with project managers.

Q5: What kinds of projects suite Agile practices? Our final question aimed at finding out when to use Agile practices. The question assumes differences between project types as well as different phases of projects. Based on current experiences with different kinds of projects we wanted to make recommendations when to use Agile practices. The data was collected from project managers during interviews.

III. RESULTS

A. Quantitative Results

In order to obtain data to answer our questions we have conducted a number of interviews and used company internal IT systems. The answers to questions about project profitability, team performance, and customer satisfaction were given by project managers. The answers could be in the scale of the following values: 1-Poor, 2-Satisfactory, 3-Average, 4-Good, and 5-Excellent. The details of this data collection process and comments on the data validity are described in Section III-G. The obtained results have been primarily arranged according to the research questions that we have formulated in Section II. In the following sections we present the aggregated results that we have obtained so far. The individual project results are not presented in this work.

B. Q1: Do the current Agile practices benefit projects?

The results are presented in Table I. The results for different projects were grouped according to the methodology used. Then for different groups the average results were calculated. Also the table contains the number of projects belonging to certain category.

Table I: Projects grouped by the methodology used

Number of projects	Methodology	Customer satisfaction	Profitability	Team performance
4	Iterative	4.5	2.5	3.5
8	Scrum	3.75	4.13	4.13
6	WF	3.5	2.83	3.83
18	Grand Total	3.83	3.33	3.89

The results in Table I show that there were 8 Scrum, 4 Iterative, and 6 traditional (Waterfall-like; short *WF*) projects. The *Customer Satisfaction* result 3.75 obtained by Scrum projects was just below the total average Customer Satisfaction 3.83 and below the result of Iterative projects 4.5. Therefore, it seems that Customer satisfaction should be addressed better in the Scrum projects. The *Profitability* in Scrum projects was 4.13, while Iterative projects had profitability at the level of 2.5 and the Traditional projects at the level of 2.83. In this case Scrum projects seem to perform much better than non-Scrum projects. Finally, the *Team Performance* in Scrum projects was 4.13. The Iterative and Traditional projects had Team Performance at levels of 3.5 and 3.83 respectively. These results for Scrum projects are above the total average result of 3.89.

Answer: The obtained results indicate that Profitability and Team Performance are better in the case of Scrum projects. The strongest positive difference is visible especially in the case of Profitability, which also is best measurable. Customer satisfaction, on the other hand, was below the total average score. To sum up, the answer to the Question 1 would be yes, Agile practices (i.e. Scrum) seem to benefit our projects in which they were used.

C. Q2: Does the customer direct involvement in the project benefit project success?

The data to answer this question was prepared by grouping projects that had customer involvement and those that did not have it. Furthermore, we also grouped the projects based on the methodology used to select potential differences between the methodologies. Table II gathers the obtained results.

The obtained results show that there were three types of

Table II: Customer involvement in projects

Number of projects	Meth.	Customer satisfaction	Profit.	Team perf.	Customer involvement
3	Iterative	4.33	2.33	3.33	
1	Scrum	1	4	4	
3	WF	4	2.33	4	
7		3.71	2.57	3.71	No Result
1	Iterative	5	3	4	
3	Scrum	4.67	4.67	4.33	
1	WF	4	3	4	
5		4.6	4	4.2	Partial Result
4	Scrum	3.75	3.75	4	
2	WF	2.5	3.5	3.5	
6		3.33	3.67	3.83	Yes Result
18		3.83	3.33	3.89	Grand Total

Table III: Communication in different project types

No. proj.	Comm. factor	Methodology	PM time
4	22.76%	Iterative Result	10.22%
8	28.12%	Scrum Result	12.42%
6	26.51%	WF Result	9.40%
18	26.39%	Grand Total	10.92%

customer involvement - *No* involvement, *Partial*, and full customer involvement marked as *Yes*. Most of the projects had some customer involvement (11 projects). Projects that did not have any customer involvement were mostly traditional types of projects. The only Scrum project that did not have any customer involvement also received the lowest Customer Satisfaction score. On the other had projects that had some customer involvement were in most of the cases Scrum projects.

The *Customer Satisfaction* for projects with no, partial and full customer involvement was 3.71, 4.6, and 3.33, respectively. It seems that partial customer involvement worked best for the investigated projects. The results encourage us to establish better measurement methods of customer satisfaction, and at the same time pay attention to the customer role in the projects.

The *Profitability* for the projects that did not have any customer involvement was 2.57, while for those with partial and full involvement the profitability was 4 and 3.67, respectively. Profitability was better in projects that had customer involvement and the best profitability level was achieved in projects with partial customer involvement. Additionally, the Scrum projects with partial customer involvement had the highest profitability level.

Finally, the *Team Performance* level achieved in projects with no, partial, and full customer involvement was 3.71, 4.2, and 3.83, respectively. Again it seems that the best suitable for the projects was partial customer involvement, and the lowest score was for the project without any customer involvement.

Answer: Based on the collected data we can say that the customer involvement in the investigated projects was beneficial, which answers the Question 2. However, the decision on to what extent the customer should be involved in the project should be adjusted to the project needs.

D. Q3: Is the communication different in Agile projects?

In order to investigate the communication efforts in relation to the project methodology used, we measured the time spent for communication to the total time of a project. The communication tasks included project meetings, project management, review meetings, and any other project-specific tasks that were regarded by the project manager as tasks that spread knowledge in the team. This metric is referred to as *Communication Factor*, or short *Comm. factor*. Additionally, we also included the time reported as the time spent for the project management. The obtained results are presented in Table III.

The *Communication Factor* for Scrum projects was 28.12%. The Iterative and Waterfall projects had the

communication factor at levels of 22.76% and 26.51%, respectively. There is a noticeable difference between the Iterative and Scrum projects as for the time spent for communication. The *Project Management time* was the highest for the Scrum projects 12.42%. While Iterative and Waterfall projects required project management time of 10.22% and 9.40% respectively. Also in this case the Scrum projects required a bit more than average management time.

Answer: After analysing the communication-related data from the investigated projects there are visible differences in communication patterns in Scrum and non-Scrum projects. Therefore, the answer to Question 3 would be yes, the investigated Scrum projects had higher communication factor than non-Scrum projects. However, the differences were not significant, which also indicates that there is no additional cost in Scrum projects due to the communication.

E. Q4: How to adapt Agile practices to suite commercial needs?

In order to answer this question we have gathered a list of different Agile elements used in some Solita projects. We also gathered some remarks that were mentioned in project manager interviews and that add some more information on how and why a particular practice was used. However, in this paper we are only able to present an indicative list of the Agile elements that were found relevant by interviewees. The Agile elements that have been commonly used are: *Scrum meetings*, *Product Backlog*, *Sprint length*, *Planning sessions*, *Sprint demonstration and retrospective*, *Customer involvement*, *Continuous integration*, *Automated tests*, *Collaboration tools*, and *Documentation*.

Answer: Based on the collected list of Agile practices used at Solita and the additional comments we have collected during the interviews we can recommend usage of all the above mentioned practices.

F. Q5: What kinds of projects suite Agile practices?

This is another question that is being answered based on our observations and comments received during interviews. The data gathered to date about different projects allows us to formulate some notes on Agile practices in different projects.

Agile practices can be successfully used in different project types, where projects differ by the technology used, team organisation, as well as the contract model governing the project. For example, we have observed successful software projects and data-warehouse projects. Agility is particularly suitable whenever requirements are likely to change.

Agile approach is especially suitable for projects with Time and Material contract models, where the clients decide how much effort is spent in the project and also they decide when the project is completed. However, Agile methodology has been successfully used in projects with fixed contract and scope. In which case Agile practices were used more internally to implement the project.

The project stage where Agile practices typically were used was the implementation stage. The implementation is the most natural stage of a project where Agile practices can be used, as then the core of the project is realised and all team members are involved. However, some Agile practices can also be used at project preparation or maintenance, especially those practices that improve communication or ensure code quality.

Our average team size was about 6 members, but teams from 3 members already seem feasible. On the other hand, teams larger than 10 members may require special arrangements (e.g., Scrum of Scrums meetings), but we are not addressing that area as we have limited experience with such projects.

G. Data Collection and Validity

We have collected the data using different sources. The main source was the time tracking system that allows for generating reports on time spent for individual projects at a single task level. Also the system distinguishes development tasks from management, supportive, or other tasks. That task classification was utilised to generate directly reports for projects. Additionally, the system keeps track of invoices related to projects, therefore the system allows for gaining insight into profitability of projects. In addition to the quantitative data obtained from the IT system, we have carried out interviews with project managers of the individual projects. The interviews were conducted according to a uniform template. Finally, the free text feedback we have received is used to provide the most prominent remarks that correlate to data analysis results, as they give additional insight into the projects.

The risks that might affect data validity were also recorded and they were already mentioned during results analysis. One weakness of the data is the fact that the customer satisfaction was calculated only based on the informal feedback received by project managers. Therefore the accuracy of such feedback can differ between projects and clients. In the future we should be able to obtain more accurate data, as Solita will start collecting such feedback in a uniform way.

Another aspect of the data validity is the time tracking system usage, which has not been uniform between the projects. We limited this risk by asking each project manager for interpretation of data, for example, to recognise communication tasks. Finally, the fact that the data to this research has been gathered by the internals of the company on one hand gave insight into any aspects of project realisation. But on the other hand, the insider point of view might affect the data interpretation. In this case we hope that combination of internals and academy representatives during final results analysis helped to achieve an objective point of view.

IV. RECOMMENDATIONS

Based on the collected data and answers to our questions the following initial recommendations for usage of Agile practices can be formulated.

Agile practices should be used whenever possible, as they seem to benefit projects. Agile project organisation can be seen as the default practice, unless some project-specific constraints prevent that, e.g., methodology imposed by the customer. Also whenever possible customer should be involved in the project to provide feedback. Agile projects can use tools and practices as listed in Section III-E.

It does not seem to be necessary to allocate more resources to the project due to communication or management. Also there is no need to increase effort estimates of the whole projects (e.g., done when offer for the customer is prepared) only due to the usage of Agile methodology.

Whenever possible Agile projects should be offered under time and material type of contracts, which give the client the freedom to decide the scope of the project but also they require the client involvement.

Agile projects still require some preparation and planning at least at a high level of goals, general architecture of a system, and effort estimates and scheduling.

V. CONCLUSIONS

We have conducted a study to evaluate Agile practices in a software service company. Using the GQM methodology, we have analysed a number of projects and answered a set of questions using specific metrics. Those answers allowed us to formulate a number of recommendations, which aim at helping to further improve the realisation of projects at Solita. Additionally, the results of this research can be utilised by other companies in a similar context. Furthermore, the results and recommendations can be used as a reference for further studies. From Solita's perspective, we also aim to use the results to verify how future Agile projects compare with those implemented in the past. This is essential in order to continue the efforts to provide high quality software and optimise resources. As future work, we plan to extend this study with other ingredients such as company sites and subcontractors.

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